Southeast University

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DATA AND SECURITY

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1. Introduction

In the modern digital era, data has become the most valuable asset for organizations worldwide. From intellectual property and financial records to personally identifiable information (PII) and protected health information (PHI), the digitization of assets has led to unprecedented operational efficiency. However, this paradigm shift has concurrently introduced a vast and evolving attack surface.

Data & security refers to the protective digital privacy measures that are applied to prevent unauthorized access to computers, databases, and websites. It also protects data from corruption. Data security is an essential aspect of IT for organizations of every size and type. The core objective of data security is to ensure the continuous availability, integrity, and confidentiality of data throughout its lifecycle. This report examines the growing challenges of data security in today's rapidly evolving digital environment, where increasing reliance on cloud systems and interconnected technologies has significantly expanded the cyber threat landscape. It focuses on key data protection mechanisms such as encryption, authentication, and authorization, which form the foundation of secure digital systems. The study is based on the CIA Triad—Confidentiality, Integrity, and Availability—as a core framework for evaluating security effectiveness. It also explores the shift from traditional security models to modern approaches such as Zero Trust Architecture, which emphasizes continuous verification of users and systems. By analyzing contemporary cybersecurity threats and defensive technologies, the report highlights the importance of implementing robust security frameworks to reduce risks, protect sensitive data, and ensure the reliability of digital infrastructures. The findings emphasize that effective data security is essential for maintaining trust and stability in the modern interconnected digital world.

2. Main Discussion

2.1. Data and Security

In the contemporary digital landscape, data has evolved far beyond its traditional role as processed information to become the fundamental currency of the global economy. As the raw architecture of modern life, data is constantly generated, exchanged, and archived across an increasingly complex web of digital systems. From personal financial records and medical histories to the strategic intellectual property of multinational corporations, the sheer volume and velocity of information being utilized today have reached unprecedented levels. This rapid digital expansion, while fueling innovation and connectivity, has simultaneously created a critical dependency on the systems that store and manage this collective intelligence. As this dependency grows, the discipline of data security has emerged as the essential shield for the digital world. Data security is defined as a comprehensive framework encompassing advanced technologies, rigorous organizational policies, and proactive defense practices designed to protect information from unauthorized access, exfiltration, or corruption. The ultimate objective of these security measures is to uphold the integrity of the CIA Triad, a foundational model that balances the confidentiality of sensitive records, the accuracy of the data itself, and the reliable availability of systems for authorized users. Without these three pillars, the trust required for digital interactions would effectively vanish. The integration of robust security measures is now a universal requirement across every major sector of society. In the financial industry, sophisticated encryption and authentication protocols are the backbone of global banking, ensuring the sanctity of transactions and personal wealth. Similarly, the healthcare sector relies on these protections to maintain patient privacy and the operational uptime of life-critical medical devices. Beyond these, educational institutions, government bodies, and private enterprises utilize data security to safeguard national infrastructure, protect student records, and preserve the consumer trust that is vital for long-term economic sustainability. However, the importance of this protection has never been more urgent due to the escalating sophistication of the modern threat landscape. The rise of automated cyber threats, such as AI-enhanced phishing, complex ransomware networks, and global data breaches, has shifted security from a technical luxury to a non-negotiable priority. A failure to implement adequate defenses can lead to catastrophic consequences, ranging from devastating financial losses and legal litigation to the permanent destruction of an organization's reputation. In an era where a single vulnerability can be exploited in seconds, ensuring a resilient security posture is the only way to navigate the risks of an interconnected environment. This report provides a strategic examination of the global cybersecurity landscape as it stands in 2026. By analyzing various threat categories and their associated financial impacts, the following discussion identifies the core technologies and methodologies required to defend against modern adversaries. Through this comprehensive evaluation, the report highlights the critical necessity of adopting adaptive security frameworks, such as the Zero Trust model, to ensure that the digital future remains both innovative and secure.

2.2. Data Classification and Sensitivity

Before securing data, an organization must understand what data it possesses. Data classification is the process of organizing data into categories for its most effective and efficient use. A well-planned data classification system makes essential data easy to find and retrieve. More importantly, it dictates the level of security controls applied to specific datasets.

Common classifications include:

- **Public:** Information freely available to anyone (e.g., marketing materials).
- **Internal:** Information restricted to employees (e.g., company directories).
- **Confidential:** Sensitive business data (e.g., vendor contracts).
- **Restricted/Secret:** Highly sensitive data requiring maximum security (e.g., trade secrets, user passwords, credit card numbers).

2.3. Identity and Access Management (IAM)

Identity and Access Management is the framework of policies and technologies ensuring that the right users have the appropriate access to technology resources. IAM systems not only identify, authenticate, and authorize individuals but also hardware and software applications. Core components include Role-Based Access Control (RBAC), Multi-Factor Authentication (MFA), and the Principle of Least Privilege, which ensures users are granted only the permissions absolutely necessary to perform their jobs.

2.4. Encryption and Data Protection

Encryption is the cryptographic transformation of data into a format unreadable by anyone without a secret decryption key.

- **Data at Rest:** Protecting data stored on physical or virtual mediums (e.g., AES-256 for hard drives and databases).
- **Data in Transit:** Protecting data moving across networks (e.g., TLS/HTTPS for web traffic).

Proper key management (using Hardware Security Modules or Cloud KMS) is critical; if encryption keys are compromised, the encryption itself is rendered useless.

3. Increase of Data Usage and Lack of Data Security

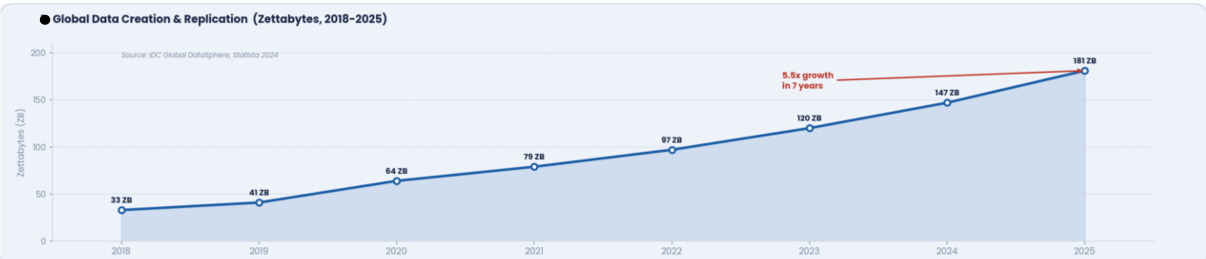

Figure 1: Usages of Data

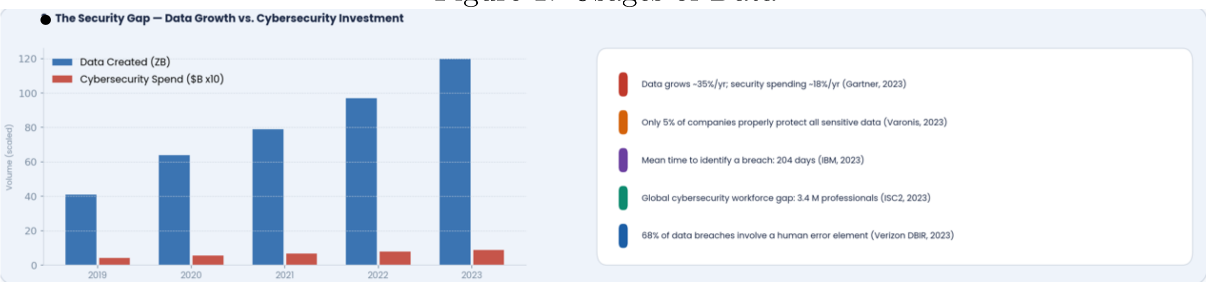

Figure 2: Lack of Data Security

4. Data Table and Chart

To understand the practical implications of data security failures, we must examine industry trends regarding data breaches over the past five years.

4.1. Data Table

The following table presents data on the number of Threat Category, Primary Technologies, Incidents and impact on financial cost associated.

Threat Category	**Primary Technologies**	**Incidents**	**Impact (M$)**
Phishing	Email Security, MFA, Training	5.2M	4.76M$
Password Attacks	MFA, IAM, Password Managers	2.8M	3.9M$
DDoS	WAF, CDN, Traffic Filtering	1.5M	2.1M$
AI Attacks	AI Detection, Analytics, SOAR	1.2M	7.8M$
Cloud/Exposure	CSPM, DLP, CASB	950K	4.5M$
IoT Attacks	IoT Security, Segmentation	900K	3.2M$
Ransomware 2.0	EDR, XDR, Backup Systems	750K	5.13M$
Insider Threats	UEBA, Zero Trust, IAM	300K	4.90M$
Supply Chain	SBOM, Vendor Risk Mgmt	120K	4.46M$

4.2. Chart from Excel

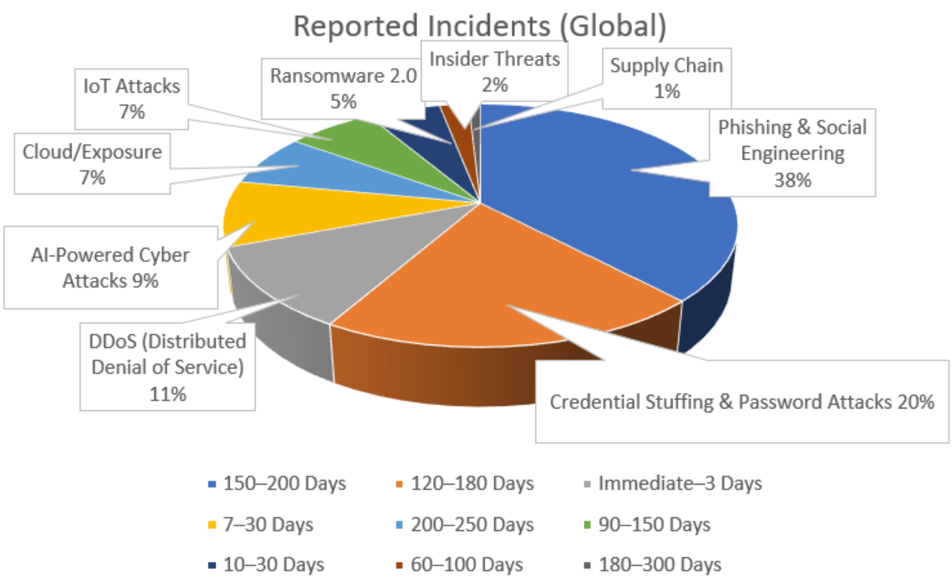

Figure 3: Average Detection Time (Days) of Reported Incident (Global)

5. Explanation of the Chart

As illustrated in Data Table and Figure 4, there is a clear and alarming upward trajectory in both the frequency of data breaches and the financial devastation they cause.

Almost 60% of all incidents come down to just two things: Phishing (38%) and Credential Stuffing (20%). This confirms that the biggest vulnerability in tech is still the human element. Whether it's clicking a bad link or reusing an old password, these "low-tech" entries are by far the most successful ways for hackers to get in.

- The Big Problems take the longest to find: Phishing (the biggest blue slice) corresponds to a 150–200 day detection window. That means a hacker could be sitting in an email system for half a year before anyone notices.
- DDoS attacks (the grey slice at 11%) are discovered almost immediately, simply because they crash the site. It's hard to ignore a house on fire, but it's easy to miss a termite infestation.
- AI-Powered Attacks (9%) and Ransomware 2.0 (5%) may seem like smaller slices, but they represent the "quality over quantity" shift. Even though they happen less frequently than phishing, they often move faster (7–30 days) and do significantly more damage to a company's bank account and reputation..

Cloud Misconfigurations and Supply Chain Attacks. While they only make up a tiny fraction of total incidents (7% and 1% respectively), they have some of the longest detection times—up to 300 days for supply chain breaches. These are the "silent killers" of the cyber world; by the time you find them, the data is usually long gone.

6. AI Usage Table

Tool Used	Purpose	Example Prompt (Short)	Output Used	Modification Done	Verification Method	Who Used It
Chat GPT	Understanding concept	"What is Data & Security?"	Used for explanation	Simplified & rewritten	Checked with website	2026000000004
Gemini	Finding sub topic	"Give some topic of Data & Security" "Make a diagram of Data & Security"	Used for presention topic and report making topic	Simplified & rewritten	Checked with website	2026000000007
Claude	Topic explanation and data collection	"Details explanation of specific topic"	Used in report making and excel sheet	Simplified & rewritten	Checked with website	2026000000017
Github Copilot Pro	Solve repository, readme file error	"Help me to solve readme file issue like table making and organising readme file structure"	Used in github repository	Simplified & rewritten	Checked with website	2026000000004

Figure 4: AI Usage Table

7. Team Member Contribution

Name	ID	Contribution
MD. MAHFUZ SARKER MAHI	2026000000004	Editing & Formation
FARZANA AHMED	2026000000007	Gather Information Diagram Making
TASPRIA CHOWDHURY ANISHA	2026000000017	Topic Overview Overview of Repository

8. Conclusion

Data security is no longer an isolated IT function; it is a fundamental business imperative. As the data presented in this report indicates, Cyber adversaries are becoming more prolific, and the financial consequences of inadequate security are growing more severe.

While data is the most valuable asset of the digital age, its vulnerability remains our greatest risk. As technology continues to evolve, so do the complexities of cyber threats. Therefore, it is no longer enough to simply utilize data; we must establish an impenetrable shield of security through robust encryption, strong passwords, and constant personal awareness. Ultimately, the synergy between the free flow of information and flawless security is what will ensure a resilient and secure digital future.

9. References

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